

The Loom

An identification game about individual treatment effects

Companion note to the standalone HTML game

16 September 2026

Abstract. The Loom turns an identification problem into a constructive game. Players observe exact income distributions with and without a programme and connect their entries with threads. Each weave describes a possible joint distribution of potential outcomes. Although every weave has the same average treatment effect, different weaves can imply different shares benefiting, losing, or gaining substantially. Players submit two witness weaves attaining the smallest and largest possible shares. The game distinguishes point identification, sharp partial identification, additional observed information, and additional assumptions. Its referee checks all admissible pairings exactly.

1 Mechanics and an opening example

The player works for a ministry whose records establish the two marginal income distributions. A commission asks for bounds on a particular population share. The board displays untreated incomes on the left and treated incomes on the right. A thread represents the two potential incomes of a hypothetical person, or an equally weighted population group. These links are proposed explanations of the records. They are not observed identities or matches between particular participants in two experimental arms.

Tapping two right endpoints, or dragging one onto another, exchanges their threads. Every endpoint must be used once. The board displays each hypothetical income change and the share meeting the commission’s criterion. The player keeps one weave as a lower-bound witness and another as an upper-bound witness, then submits the report. A correct submission must attain both sharp endpoints. If it fails, the referee explains which endpoint can be improved; the player may request a counterexample. Cases completed with this assistance are marked separately.

In the opening commission, each entry carries one quarter of the population. Untreated incomes are $(0, 1, 2, 3)$ and treated incomes are $(1, 2, 3, 4)$. Table 1 gives three different admissible weaves. Each column on the right contains exactly the same treated incomes.

Table 1: Three possible worlds with the same observed distributions

Without programme	World A	World B	World C
0	1	4	3
1	2	1	4
2	3	2	1
3	4	3	2
Share benefiting	100%	25%	50%
Share losing	0%	0%	50%
Average effect	1	1	1

The sharp benefit bounds are $[1/4, 1]$. At least one group must gain because the total effect is positive; World B attains this minimum and World A the maximum. World C shows that the positive average can coexist with losses for half the population.

2 What is identified

Let the two equally weighted lists be $a = (a_1, \dots, a_n)$ and $b = (b_1, \dots, b_n)$, with $n \in \{4, 6, 8\}$ in the implementation. Their distributions are known exactly. The game therefore studies identification given the marginal laws; it does not simulate the sampling error of a finite experiment. A possible weave is a permutation π of $\{1, \dots, n\}$, giving individual effects

$$d_i(\pi) = b_{\pi(i)} - a_i.$$

Let $\mathcal{A}$ be the nonempty set of permutations consistent with the stated evidence and assumptions. Initially, $\mathcal{A}$ contains every permutation. For a commission with criterion g , the target is

$$\theta_g(\pi) = \frac{1}{n} \sum_{i=1}^n g(d_i(\pi)), \quad g(d) = \mathbf{1}\{d > 0\}, \quad \mathbf{1}\{d < 0\}, \quad \text{or} \quad \mathbf{1}\{d \geq c\}. \quad (1)$$

These criteria describe benefit, harm, and a gain of at least c income units. The boundary conventions matter: zero is neither a benefit nor a loss; a gain exactly equal to c meets the last criterion.

Proposition 1 (The average effect). *The average treatment effect is point identified from the two marginal distributions, for every nonempty admissible set:*

$$\mu = \frac{1}{n} \sum_i d_i(\pi) = \frac{1}{n} \sum_j b_j - \frac{1}{n} \sum_i a_i. \quad (2)$$

Proof. A permutation uses every treated income exactly once, so $\sum_i b_{\pi(i)} = \sum_j b_j$. The expression is independent of π . $\square$

Proposition 2 (Sharp bounds and point identification). *The sharp lower and upper bounds on a commission's target are*

$$L_g = \min_{\pi \in \mathcal{A}} \theta_g(\pi), \quad U_g = \max_{\pi \in \mathcal{A}} \theta_g(\pi). \quad (3)$$

The target is point identified if and only if $L_g = U_g$.

Proof. The finite, nonempty set $\mathcal{A}$ contains a minimiser and a maximiser. Every admissible target value lies between their values. Raising the lower endpoint or lowering the upper endpoint excludes one of these admissible worlds. The target has a unique value precisely when the endpoints coincide. $\square$

In the game's finite representation, the exact identified set is $\{\theta_g(\pi) : \pi \in \mathcal{A}\}$. It need not contain every real number between the bounds. The submitted interval reports its sharp endpoints, not a claim that every interior value occurs on the finite board.

Identification is specific to the target. If $\min_j b_j > \max_i a_i$, every person benefits under every weave: $\theta_+ = 1$. Yet many pairings and distributions of effect sizes remain possible. Conversely, equal marginal distributions need not imply zero individual effects. For $a = b = (0, 1, 2, 3)$, the benefit bounds are $[0, 3/4]$: the identity weave attains zero and a cyclic weave attains three quarters. Strict gains for everyone would contradict the known zero sum.

3 What additional information changes

The game explicitly distinguishes restrictions supplied by the records from restrictions supplied by an assumption. Each changes $\mathcal{A}$ and may change the answer. Restricting the admissible set can only weakly raise a lower bound and weakly lower an upper bound, provided the restricted set is nonempty. It need not make either change strict.

Observed district labels. Some commissions reveal a characteristic fixed before the programme, together with both outcome distributions within each category. If the labels are x_i and z_j , admissible weaves must satisfy $x_i = z_{\pi(i)}$. For example, consider $a = (0, 1, 2, 3, 4, 5)$ and $b = (1, 2, 3, 4, 5, 6)$. Without labels, the benefit bounds are $[1/6, 1]$. If the first three entries in each column belong to North and the last three to South, every admissible weave preserves these districts. Each district has a positive average effect, requiring at least one beneficiary. The lower bound rises to $2/6$, while the upper bound remains one.

District information can also identify a share exactly. If every treated income exceeds every untreated income within North, while every treated income is lower within South, the share benefiting equals North's population share. Individual effect sizes can remain unknown. These are consequences of additional observations, not of an assumption that people preserve their ranks.

No harm. Other commissions explicitly assume

$$b_{\pi(i)} \geq a_i \quad \text{for every } i. \quad (4)$$

The board rejects a swap that violates this restriction. In the opening example, no harm does not narrow the benefit bounds: both Worlds A and B remain possible. It does narrow the upper bound on the share gaining at least two. Without the assumption, three quarters can do so; with it, at most half can. The total gain is four, so three gains of at least two would require a compensating loss. Two gains of two and two zero effects attain the restricted maximum.

No harm becomes decisive when the mean effect is zero. All $d_i \geq 0$ and $\sum_i d_i = 0$ imply $d_i = 0$ for every i . The benefit share, harm share, and entire distribution of effects are then identified. The assumption must remain visible: the unchanged average alone does not establish this result.

Preserved ranks. A further commission assumes that no two people reverse their income ordering:

$$(a_i - a_k)(b_{\pi(i)} - b_{\pi(k)}) \geq 0 \quad \text{for all } i, k. \quad (5)$$

Ties are allowed. This produces the ordered, or comonotone, coupling. In the game, $a = (0, 1, 2, 3, 4, 5)$ and $b = (1, 1, 3, 3, 5, 5)$ yield effects $(1, 0, 1, 0, 1, 0)$ and an identified benefit share of one half. Permuting equal-valued endpoints changes no substantive outcome. Rank preservation is not an implication of random assignment; it supplies the missing link between the two regimes.

More precision and more information. Repeating an experiment that supplies only the same marginal laws cannot resolve their missing dependence once those laws are already known exactly. Information about baseline characteristics may help by furnishing conditional laws. Structural restrictions may help by reducing admissible couplings. The game separates these two sources of progress and leaves sampling uncertainty to an extension.

4 Adjudication and the scope of the model

The HTML implements a complete search through admissible permutations. It prunes pairings that violate district labels, no harm, or preserved ranks, then evaluates the target at every complete admissible weave. With at most eight endpoints per column, there are no more than $8! = 40,320$ permutations. The referee therefore uses exact integer counts, with no simulation, numerical optimiser, search-depth cap, or tolerance in its verdict. Percentages are rounded for display only.

A submission succeeds when the saved lower witness attains the true minimum and the upper witness attains the true maximum. Equal endpoints receive the verdict *Point identified*; unequal sharp endpoints receive *Sharp bounds*. A rejected submission can be met by another admissible weave outside the submitted bounds. Finding two valid witnesses establishes attainability; the exhaustive referee additionally establishes that neither can be improved. No secret true weave is generated, and matching an arbitrary hidden truth is never the objective.

Table 2: The ten introductory commissions and their exact answers

Case	Target	Additional restriction	Sharp bounds
1	Benefit	None	$[1/4, 1]$
2	Benefit	None; equal marginals	$[0, 3/4]$
3	Harm	None	$[0, 2/3]$
4	Benefit	None; separated supports	$[1, 1]$
5	Benefit	Observed districts	$[1/3, 1]$
6	Benefit	Observed districts	$[1/2, 1/2]$
7	Gain of at least 2	Assumed no harm	$[0, 1/2]$
8	Benefit	Assumed no harm; zero mean	$[0, 0]$
9	Benefit	Assumed preserved ranks	$[1/2, 1/2]$
10	Gain of at least 3	None	$[0, 3/4]$

Why equally weighted threads suffice for the bounds. In an unrestricted population interpretation, mass may be split across several potential partners. Write q_{ij} for the joint mass on (a_i, b_j) . Its row and column sums are $1/n$, and the target is the linear functional $\sum_{ij} q_{ij}g(b_j - a_i)$. The matrix nq is doubly stochastic and is a convex combination of permutation matrices. Consequently, its target value is a convex combination of values attained by the game's weaves; the same minimum and maximum apply. District and no-harm restrictions simply forbid particular edges, and this conclusion continues to hold: a zero entry can occur only if every positively weighted permutation also avoids that entry. Preserved ranks instead fixes the ordered joint law directly. Thus indivisible threads do not artificially tighten the bounds for the restrictions implemented here. With divisible mass, intermediate target values between the unrestricted or edge-restricted endpoints are attainable by mixtures.

Use and limitations. The file opens directly in a modern browser and works offline. It contains ten introductory commissions, generated free commissions, undo, saved and restorable witness weaves, keyboard controls, and optional referee counterexamples. Completion records are stored locally when the browser permits it. All populations have exact discrete marginal distributions and stable potential outcomes. Treatment assignment, finite-sample inference, interference and the credibility of substantive assumptions are outside the simulated environment. The educational task is to distinguish an identified average, an identified target share, and an identified joint distribution, and to recognise a sharp range as a completed finding.